Tree

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1 Tree

I Tree

i Tree

A tree is an abstract data type of a set of nodes. A relationship between two nodes i, j is defined with

- i is called a child of j and
- j is called a parent of i; and
- a node can have at most one parent.

If not empty, a tree is recursively defined as a distinguished node r called root and zero or more nonempty trees called subtrees, where the root of each substree is a child of r.

ii Terminology

- Siblings: Nodes sharing the same parent are called siblings. Children of a node are defined with a left to right order.
- Leaf: A node with no children is called a leaf, external node, or exterior node.
- Internal node: A node that is not a leaf is called an internal node or interior node.
- Depth: The number of edges from the root to a node is called the depth of the node, while the root has depth 0.
- Height: The height of a leaf is defined as 0. The height of an internal node is defined as the maximum height of its children plus 1. The maximum depth of the nodes, i.e., the height of its root, is called the height or depth of the tree and denoted h. The height or depth of an empty tree is defined as -1. Some define heights of nodes and trees as the definition here plus 1, which is not used here.
- Ancestors: The ancestors of a node is recursively defined as itself for root and itself and the ancestors of its parent for non-root nodes.
- Descendants: The descendants of a node is recursively defined as itself for leaves and itself and the descendants of its children for internal nodes.
- Level: Nodes of the same depth are called a level. Levels are ordered by depth. The left-to-right order of nodes in a level is defined recursively to, if they are siblings, follow that of siblings; otherwise follow that of their parents.
- Subtree: A tree consisting of all descendants of a node is called a subtree. The left-to-right order of subtrees whose roots are of the same depth are defined as

that of their roots. A subtree whose root is a child of a node is called a subtree of the node or rooted at the node.

- The umber of nodes in a tree is denoted n.

iii Tree representation

A representation of tree where each pair of parent and child is connected with a edge.

iv Left child, right sibling representation

A representation of tree where each internal node and its leftmost child is connected with a edge, and each sibling that is not the leftmost one is called to the sibling at its left.

v Skewed

A tree whose every level contains only one node is called skewed.

II n-ary tree

i n-ary tree

A tree in which no node can have more than n children is called an n-ary tree. A child is indexed into one and only one integer in $[1, n]$ even if its parent has less than n children.

ii Full, proper, or plane

An n-ary tree where each internal node has exactly n children is called full, proper, or plane.

iii Filled

A level of depth $k \in \mathbb{N}$ of an n-ary tree is called filled iff each node in the level of depth $k - 1$ has exactly n children. The level of depth 0 is called filled iff the tree is nonempty.

iv Complete

An n-ary tree where every level, except possibly the last, is filled and whose nodes in the last level are as far left as possible, is called complete.

A complete n-ary tree is indexed from top to bottom and then from left to right, with the root as node 1.

v Perfect

A complete n-ary tree where all leaves are of the same depth is called perfect.

A perfect n-ary tree with height k has $\sum_{j=0}^{k} n^j$ nodes.

III Binary tree

i Binary tree

In a binary tree, the two possible children of a node x are called left child, denoted $x.\text{left}$, and right child, denoted $x.\text{right}$. A child is either left or right but not both even if it has no sibling. The left and right subtrees of a node x are denoted $\text{left}(x)$ and $\text{right}(x)$ respectively.

ii Complete binary tree

For a complete binary tree with n nodes:

- Parent of node i is node $\left\lfloor \frac{i}{2} \right\rfloor$ for all integers $i > 1$.
- The left child of node i is $2i + 1$ for all integers $i \leq \frac{n-1}{2}$.
- The right child of node i is $2i$ for all integers $i \leq \frac{n}{2}$.

iii Perfect binary tree

A perfect binary tree with height k has $2^{k+1} - 1$ nodes.

iv Balanced tree

A type of binary tree with height $O(\log n)$ where n is the number of nodes is called balanced.

v Balance factor (bf)

The height of the left subtree minus the height of the right subtree of a node in a binary tree is called the balance factor of the node.

vi Left-skewed and right-skewed

A binary tree whose every internal node has left child and does not have right child is called left-skewed.

A binary tree whose every internal node has right child and does not have left child is called right-skewed.

IV Tree traversal

They are defined recursively. Suppose the root is r.

i Preorder traversal

First visit r, and then visit the subtrees of r from left to right.

ii Postorder traversal

Visit the subtrees of r from left to right, and finally visit r.

iii Inorder traversal (binary tree)

First visit the left subtree of r, and then visit r, and then visit the right subtree of r.

iv Full binary tree with distinct values can be uniquely determined by its preorder and postorder traversals

A full binary tree with distinct values can be uniquely determined by its preorder and postorder traversals.

Proof. The following C code provides an algorithm to reconstruct the tree recursively:

```c
#include <stdlib.h>

typedef struct Node {
  int data;
  struct Node *left;
  struct Node *right;
} Node;

Node *createNode(int data) {
  Node *n = (Node *)malloc(sizeof(Node));
  n->data = data;
  n->left = NULL;
  n->right = NULL;
  return n;
}

/*
 * Reconstruct a full binary tree from:
 *   P - preorder array (size num)
 *   Q - postorder array (size num)
 *   distinct values assumed
 */
Node *construct(int *P, int *Q, int num) {
  if (num <= 0)
    return NULL;
  if (num == 1)
    return createNode(P[0]);
  Node *root = createNode(P[0]);
  int leftRootVal = P[1];
  int k = 0;
  while (Q[k] != leftRootVal)
```

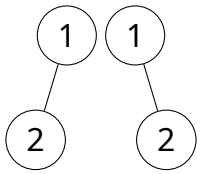
```
    k++;
  int leftSize = k + 1;
  int rightSize = num - 1 - leftSize;
  root->left = construct(P + 1, Q, leftSize);
  root->right = construct(P + 1 + leftSize, Q + leftSize, rightSize);
  return root;
}
```

□

v Binary tree with distinct values cannot be uniquely determined by its preorder and postorder traversals

The statement that a binary tree with distinct values can be uniquely determined by its preorder and postorder traversals is false.

Proof. No. A counterexample is the following two binary trees



Both trees have preorder traversal 1, 2, and postorder traversal 2, 1, but they are different. □

vi Binary tree with distinct values can be uniquely determined by its preorder and inorder traversals

A binary tree with distinct values can be uniquely determined by its preorder and inorder traversals.

Proof. The following C code provides an algorithm to reconstruct the tree recursively:

```c
#include <stdlib.h>

typedef struct Node {
  int data;
  struct Node *left;
  struct Node *right;
} Node;

Node *createNode(int data) {
  Node *n = (Node *)malloc(sizeof(Node));
  n->data = data;
  n->left = NULL;
  n->right = NULL;
  return n;
}

/*
 * Reconstruct a binary tree from:
```

```
 *   P - preorder array (size num)
 *   I - inorder array (size num)
 *   distinct values assumed
 */
Node *construct(int *P, int *I, int num) {
  if (num <= 0)
    return NULL;
  if (num == 1)
    return createNode(P[0]);
  int rootVal = P[0];
  Node *root = createNode(rootVal);
  int k = 0;
  while (I[k] != rootVal)
    k++;
  int leftSize = k;
  int rightSize = num - 1 - leftSize;
  root->left = construct(P + 1, I, leftSize);
  root->right = construct(P + 1 + leftSize, I + leftSize + 1, rightSize);
  return root;
}
```

□

vii Binary tree with distinct values can be uniquely determined by its postorder and inorder traversals

A binary tree with distinct values can be uniquely determined by its postorder and inorder traversals.

Proof. The following C code provides an algorithm to reconstruct the tree recursively:

```
#include <stdlib.h>

typedef struct Node {
  int data;
  struct Node *left;
  struct Node *right;
} Node;

Node *createNode(int data) {
  Node *n = (Node *)malloc(sizeof(Node));
  n->data = data;
  n->left = NULL;
  n->right = NULL;
  return n;
}

/*
 * Reconstruct a binary tree from:
 *   Q - postorder array (size num)
 *   I - inorder array (size num)
 *   distinct values assumed
 */
Node *construct(int *Q, int *I, int num) {
  if (num <= 0)
    return NULL;
  if (num == 1)
    return createNode(Q[num - 1]);
```

```
  int rootVal = Q[num - 1];
  Node *root = createNode(rootVal);
  int k = 0;
  while (I[k] != rootVal)
    k++;
  int leftSize = k;
  int rightSize = num - 1 - k;
  root->left = construct(I, Q, leftSize);
  root->right = construct(I + k + 1, Q + leftSize, rightSize);
  return root;
}
```

□

V Binary search tree (BST)

i Binary search tree (BST)

A binary search tree stores keys in the nodes of a binary tree so that searching, insertion and deletion can be done efficiently. For every node x, all the keys in its left subtree are smaller than x and all keys in its right subtree are larger than x. Thus, the values are distinct.

ii Searching

Suppose we are searching for x and the root is r.

- If $x = r$, then we are done.
- If $x < r$, then we search in the left subtree.
- If $x > r$, then we search in the right subtree.

The time complexity is $O(h)$.

iii Finding minimum and maximum

To find the node containing the smallest key, start at root and go to the left child as long as there is one, and the leaf we stop at is the smallest key. Similarly for find the largest key.

The time complexity is $O(h)$.

iv Insertion

Suppose we want to insert x. First search for it, if we found it, do noting (or update something); otherwise, if we stop at y,

- if $x > y$, insert x as the right child of y if $x > y$, and
- if $x < y$, insert x as the left child of y.

The time complexity is $O(h)$.

v Deletion

Suppose we want to delete x. First search for it, if we don't found it, do noting (or update something); otherwise,

- if it is a leaf, delete it,
- if it has one child y, adjust y as a child of the parent of x of the same position as x and delete x, and
- if it has two children, choose either of the policies:
  - inorder successor: replace the value of x with the value of the minimum element y in the right subtree of x, i.e. inorder successor, i.e., the leftmost descendant of the right child of x, which has no left child but may have right child, and
    - remove node y if it has no child, and
    - replace node y with its right child if it has one;
  - inorder predecessor: replace the value of x with the value of the maximum element y in the left subtree of x, i.e. inorder predecessor, i.e., the rightmost descendant of the left child of x, which has no right child but may have left child, and
    - remove node y if it has no child, and
    - replace node y with its left child if it has one;

The time complexity is $O(h)$.

vi (Simple) rotation

A (simple) rotation on an edge between a node p and its child x, also called rooted at or on p, with p called root and x called pivot of the rotation, is a tree operation on BST and has the following two symmetric variants:

- Right rotation: This is done when x is a left child of p. We perform the following assignments:

$$\text{left}(p) = \text{right}(x)$$

$$\text{right}(x) = p$$

- Left rotation: This is done when x is a right child of p. We perform the following assignments:

$$\text{right}(p) = \text{left}(x)$$

$$\text{left}(x) = p$$

vii Uniqueness given preorder traversal

A BST is uniquely determined by its preorder traversal.

Proof. A BST with one node is obviously unique. The first element must be root, the second element to the last element smaller than root must be the left subtree, and the rest must be the right subtree. Recursively partition into the subtrees. We uniquely obtained the BST. □

viii Uniqueness given postorder traversal

A BST is uniquely determined by its postorder traversal.

Proof. A BST with one node is obviously unique. The last element must be root, the first element to the last element smaller than root must be the left subtree, and the rest must be the right subtree. Recursively partition into the subtrees. We uniquely obtained the BST. □

ix Binary tree is BST iff inorder traversal is in ascending order

A binary tree with distinct values is a binary search tree iff its inorder traversal is sorted in ascending order.

Proof. Only if: Prove by mathematical induction. The inorder traversal of a BST with one node is sorted in ascending order. Assume the inorder traversal of any BST with fewer than n nodes of distinct values is sorted in ascending order. Consider any BST with n nodes, the inorder traversal visits the left subtree, which is smaller than root and itself sorted in ascending order since it's a BST with $n - 1$ nodes, and then root, and then the right subtree, which is larger than root and itself sorted in ascending order since it's a BST with $n - 1$ nodes. Thus, we conclude that the inorder traversal of any BST with n nodes is sorted in ascending order. By mathematical induction, the statement holds.

If: Prove by mathematical induction. A binary tree with one node is a BST. Assume that any binary tree with fewer than n nodes of distinct values whose inorder traversal is sorted in ascending order is a BST. Consider any binary tree with n nodes of distinct values and root r whose inorder traversal is sorted in ascending order. The subsequence before r is a binary tree with fewer than n nodes of distinct values whose inorder traversal is sorted in ascending order whose values of nodes are smaller than r. The subsequence after r is a binary tree with fewer than n nodes of distinct values whose inorder traversal is sorted in ascending order whose values of nodes are larger than r. Since the left subtree is a BST and smaller than root and the right subtree is a BST and larger than root, we conclude that any binary tree with n nodes of distinct values whose

inorder traversal is sorted in ascending order is a BST. By mathematical induction, the statement holds. □

x Number of distinct BST shapes

The number T_n of distinct shapes of BSTs with n nodes is:

$$T_0 = 1$$

$$T_n = \sum_{i=0}^{n-1} T_i T_{n-1-i}, \quad n \in \mathbb{N}$$

$$T_n = \frac{1}{(n+1)} \binom{2n}{n}, \quad n \in \mathbb{N}_0$$

Proof. Obviously

$$T_0 = T_1 = 1$$

Let the values be

$$k_1 < k_2 < \ldots < k_n$$

Suppose we choose k_i as root, where $1 \leq i \leq n$. The left subtree contains $k_1, \ldots, k_{i-1}$, so it has $i-1$ nodes and T_{i-1} possible combinations. The right subtree contains $k_{i+1}, \ldots, k_n$, so it has $n-1$ nodes and T_{n-i} possible combinations.

Summing over all choices gives

$$T_n = \sum_{i=1}^{n} T_{i-1} T_{n-i} = \sum_{i=0}^{n-1} T_i T_{n-1-i}$$

Using generating function:

$$C(x) = \sum_{n=0}^{\infty} T_n x^n$$

$$C(x) = 1 + xC(x)^2$$

$$C(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$$

$$C(0) = \mp \frac{1}{\sqrt{1-4x}} = T(0) = 1$$

$$C(x) = \frac{1 - \sqrt{1-4x}}{2x}$$

$$
\begin{aligned}
\sqrt{1-4x} &= \sum_{k=0}^{\infty} \binom{0.5}{k} (-4x)^k \\
&= 1 + \sum_{k=1}^{\infty} \frac{\prod_{i=0}^{k-1}(0.5-i)}{k!}(-4x)^k \\
&= 1 + \sum_{k=1}^{\infty} \frac{2^k \prod_{i=0}^{k-1}(2i-1)}{k!}x^k \\
&= 1 + \sum_{k=1}^{\infty} -\frac{2^k \prod_{i=1}^{k-1}(2i-1)}{k!}x^k \\
&= 1 + \sum_{k=1}^{\infty} -\frac{2^k \prod_{i=1}^{2k-2} i}{k! \prod_{i=1}^{k-1} 2i}x^k \\
&= 1 + \sum_{k=1}^{\infty} -\frac{2^k (2k-2)!}{k! 2^{k-1}(k-1)!}x^k \\
&= 1 + \sum_{k=1}^{\infty} -\frac{2(2k-2)!}{k!(k-1)!}x^k
\end{aligned}
$$

$$
\begin{aligned}
C(x) &= \frac{1-\sqrt{1-4x}}{2x} \\
&= \sum_{k=1}^{\infty} \frac{(2k-2)!}{k!(k-1)!}x^{k-1} \\
&= \sum_{n=0}^{\infty} \frac{(2n)!}{(n+1)!n!}x^n \\
&= \sum_{n=0}^{\infty} \frac{1}{(n+1)}\binom{2n}{n}x^n
\end{aligned}
$$

$$
T_n = \frac{1}{(n+1)}\binom{2n}{n}, \quad n \in \mathbb{N}_0
$$

□

VI AVL tree (Adelson-Velsky and Landis tree)

i AVL tree (Adelson-Velsky and Landis tree)

An AVL tree is a self-balancing binary search tree (BST) that keeps info of height, invented by Georgy Adelson-Velsky and Evgenii Landis. An AVL tree needs to comply with the property that for every node with two children in the tree, the height of its left and right subtrees differ by at most 1.

Some define heights of nodes and AVL trees as the definition used here plus 1, which is not used here.

ii Relation of height and numbers of nodes

Let the minimum number of nodes in an AVL tree of height h be N_h. Since the minimum numbers of subtrees height of the root of an AVL tree of height h are $h-1$ and $h-2$, we have:

$$N_{-1} = 0$$

$$N_0 = 1$$

$$N_h = N_{h-1} + N_{h-2} + 1, \quad h \in \mathbb{N}$$

$$N_h = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{h+3} - \left(\frac{1-\sqrt{5}}{2}\right)^{h+3}\right) - 1, \quad h+1 \in \mathbb{N}_0$$

Proof. Use characteristic equation:

$$x^2 - x - 1 = 0$$

Let

$$\varphi = \frac{1+\sqrt{5}}{2}$$

Note that

$$\frac{3+\sqrt{5}}{2} = \varphi^2 = \varphi + 1$$

$$2 + \sqrt{5} = \varphi^3 = 2\varphi + 1$$

$$\frac{7+3\sqrt{5}}{2} = \varphi^4 = 3\varphi + 2$$

$$\frac{1-\sqrt{5}}{2} = -\frac{1}{\varphi} = 1 - \varphi$$

$$\frac{3-\sqrt{5}}{2} = \frac{1}{\varphi^2} = 1 - \frac{1}{\varphi} = 2 - \varphi$$

$$x = \frac{1 \pm \sqrt{5}}{2} = \varphi, -\frac{1}{\varphi}$$

$$N_h = a\varphi^h + b\left(-\frac{1}{\varphi}\right)^h + c$$

$$N_0 = a + b + c = 1$$

$$N_1 = a\varphi - \frac{b}{\varphi} + c = 0 + 1 + 1 = 2$$

$$N_2 = a\varphi^2 + \frac{b}{\varphi^2} + c = 1 + 2 + 1 = 4$$

$$N_2 = a(\varphi + 1) - b\left(\frac{1}{\varphi} - 1\right) + c = 4$$

$$N_2 - N_1 = a + b = 4 - 2 = 2$$

$$c = N_0 - (a + b) = 1 - 2 = -1$$

$$N_1 - c = a\varphi - \frac{2-a}{\varphi} = 2 + 1 = 3$$

$$a\left(\varphi^2 + 1\right) = 3\varphi + 2$$

$$a = \frac{3\varphi + 2}{\varphi + 2} = \frac{\varphi^4}{\varphi + 2}$$

$$b = 2 - a = \frac{2-\varphi}{\varphi+2} = \frac{1}{\varphi^2(\varphi+2)}$$

$$N_h = \frac{1}{\varphi+2}\left(\varphi^{h+4} + \left(-\frac{1}{\varphi}\right)^{h+2}\right) - 1$$

$$\frac{1}{\varphi+2} = \frac{2}{5+\sqrt{5}} = \frac{1}{\sqrt{5}\varphi}$$

$$N_h = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{h+3} - \left(\frac{1-\sqrt{5}}{2}\right)^{h+3}\right) - 1$$

□

So

$$N_h = \Theta(\left(\frac{1+\sqrt{5}}{2}\right)^h)$$

$$h = \log_{\frac{1+\sqrt{5}}{2}} N_h + O(1) \approx 1.44 \log_2 N_h + O(1)$$

Let the maximum number of nodes in an AVL tree of height h be M_h. Since the maximum numbers of subtrees height of the root of an AVL tree of height h are $h-1$, we have:

$$M_{-1} = 0$$

$$M_h = 2M_{h-1} + 1, \quad h \in \mathbb{N}_0$$

$$M_h = 2^{h+1} - 1, \quad h + 1 \in \mathbb{N}_0$$

So

$$M_h \in \Theta(\log_2 M_h)$$

$$h = \log_2 M_h + O(1)$$

Let the number of nodes be n.

$$N_h \le n \le M_h$$

$$\log_2 n + O(1) \le h \le \log_{\frac{1+\sqrt{5}}{2}} n + O(1) \approx 1.44 \log_2 N_h + O(1)$$

Thus

$$h \in \Theta(\log n)$$

iii Searching and finding minimum/maximum

Searching and finding minimum/maximum in AVL tree is the same as in BST.

The worst-case and amortized time complexities are $O(h) = O(\log n)$.

iv Rotation

When insertion/deletion, the AVL property may be violated, called unbalanced. Since an insertion/deletion involves adding/deleting a single node, this can only increase/decrease the height of some subtree by 1. Thus, if the AVL property is violated at a node x, it means that the heights of left and right subtrees of it differ by exactly 2. We need to transform the tree to restore the AVL property, called rebalancing, using a single/simple rotation or a double rotation, each has two symmetric variants:

- LL case / right single rotation: Let the left child of x be y. The LL case is where the left subtree of y has height $h+1$, while the right subtree of y and the right subtree of x have height h. We perform a right rotation on the edge between x and y, i.e.,

$$\text{left}(x) = \text{right}(y)$$
$$\text{right}(y) = x$$

- RR case / left single rotation: Let the right child of x be y. The RR case is where the right tree of y has height $h+1$, while the left subtree of y and the left subtree of x have height h. We perform a left rotation on the edge between x and y, i.e.,

$$\text{right}(x) = \text{left}(y)$$
$$\text{left}(y) = x$$

- LR case / left-then-right double rotation: Let the left child of x be y and the right child of y be z. The LR case is where the right subtree of y has height $h+1$, while the left subtree of y and the right subtree of x have height h. We perform a left rotation on the edge between y and z and then a right rotation on the edge between x and z, now at the original position of y, i.e.,

$$\text{right}(y) = \text{left}(z)$$
$$\text{left}(z) = y$$
$$\text{left}(x) = \text{right}(z)$$
$$\text{right}(z) = x$$

- RL case / right-then-left double rotation: Let the right child of x be y and the left child of y be z. The RL case is where the left subtree of y has height $h+1$, while the right subtree of y and the left subtree of x have height h. We perform a right rotation on the edge between y and z and then a left rotation on the edge between x and z, now at the original position of y, i.e.,

$$\text{left}(y) = \text{right}(z)$$
$$\text{right}(z) = y$$
$$\text{right}(x) = \text{left}(z)$$
$$\text{left}(z) = x$$

v Insertion

Insertion in AVL tree involves first doing insertion as with a normal BST, and then tracing the path from the new leaf to the root: For each node x encountered, check if heights of its left and right subtrees differ by at most 1,

- if yes, proceed to the parent of x;
- if not, we say the AVL property is violated at x and perform rotations to restore it, and then break the loop since no second violation may exist.

Proof. Let x be the first node encountered when traversing upward from the newly inserted node toward the root that violates the AVL property. Since an insertion involves adding a single node, this can only increase the height of some subtree by 1. Thus, the heights of the left and right subtrees differ by exactly 2. For any of the four cases (LL, RR, LR, RL), since the inserted node must be in the subtree of height $h+1$ of y, the height of the subtree whose root is x has height $h+2$ both before insertion and after rotation. Thus, the fact that the AVL property is satisfied for all ancestors of x before insertion implies that the AVL property is satisfied for all ancestors of x after rotation. □

The worst-case and amortized time complexities are $O(h) = O(\log n)$.

vi Deletion

Deletion in AVL tree involves first doing deletion the same as in BST, and then tracing the path from

- the original parent of the deleted node if the deleted node had zero or one child, and
- the original parent of the inorder successor or inorder predecessor that was moved if the deleted node had two children,

to the root: For each node x encountered, check if heights of its left and right subtrees differ by at most 1,

- if yes, proceed to the parent of x;
- if not, we say the AVL property is violated at x and perform rotations to restore it, and then proceed to the parent of x since violation may propagate upwards.

The worst-case and amortized time complexities are $O(h) = O(\log n)$.

vii Number of distinct AVL tree shapes

The number $A_{h,n}$ of distinct shapes of AVL trees with n nodes and height h is:

$$A_{-1,0} = 1$$

$$A_{-1,n} = 0, \quad n \neq 0$$

$$A_{0,1} = 1$$

$$A_{0,n} = 0, \quad n \neq 1$$

$$A_{h,n} = \sum_{i=0}^{n-1} A_{h-1,i}A_{h-1,n-1-i} + A_{h-1,i}A_{h-2,n-1-i} + A_{h-2,i}A_{h-1,n-1-i}, \quad h \in \mathbb{N}, n \in \mathbb{N}_0$$

Proof. The root uses one node, leaving $n-1$ nodes for the two subtrees. Since the whole tree has height h, the two trees must either have heights $h-1, h-2$ or both have height $h-1$. Thus,

$$A_{h,n} = \sum_{i=0}^{n-1} A_{h-1,i}A_{h-1,n-1-i} + A_{h-1,i}A_{h-2,n-1-i} + A_{h-2,i}A_{h-1,n-1-i}$$

□

viii Number of distinct AVL tree shapes with minimum number of nodes given height

The number B_h of distinct shapes of AVL trees with height h and minimum number of nodes is:

$$B_{-1} = 1$$

$$B_0 = 1$$

$$B_h = 2B_{h-1}B_{h-2}, \quad h \in \mathbb{N}$$

$$B_h = 2^{\frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{h+2} - \left(\frac{1-\sqrt{5}}{2}\right)^{h+2}\right)-1}, \quad h+1 \in \mathbb{N}_0$$

Proof. To minimize the number of nodes, the two subtrees of root must be the AVL tree with height $h-1, h-2$ and minimum number of nodes, which have B_{h-1}, B_{h-2} distinct shapes respectively. There are two possible arrangements:

- Left subtree has height $h-1$, and right subtree has height $h-2$.
- Left subtree has height $h-2$, and right subtree has height $h-1$.

Thus,

$$B_h = 2B_{h-1}B_{h-2}$$

Let

$$C_h = \log_2 B_h$$

$$C_h = C_{h-1} + C_{h-2} + 1$$

Use characteristic equation:

$$x^2 - x - 1 = 0$$

Let

$$\varphi = \frac{1+\sqrt{5}}{2}$$

Note that

$$\frac{3+\sqrt{5}}{2} = \varphi^2 = \varphi + 1$$

$$2+\sqrt{5} = \varphi^3 = 2\varphi + 1$$

$$\frac{1-\sqrt{5}}{2} = -\frac{1}{\varphi} = 1 - \varphi$$

$$x = \frac{1 \pm \sqrt{5}}{2} = \varphi, -\frac{1}{\varphi}$$

$$C_h = a\varphi^h + b\left(-\frac{1}{\varphi}\right)^h + c$$

$$C_0 = a + b + c = \log_2 1 = 0$$

$$C_1 = a\varphi - \frac{b}{\varphi} + c = \log_2(2 * 1 * 1) = 1$$

$$C_2 = a\varphi^2 + \frac{b}{\varphi^2} + c = 0 + 1 + 1 = 2$$

$$C_2 = a(\varphi + 1) - b\left(\frac{1}{\varphi} - 1\right) + c = 2$$

$$C_2 - C_1 = a + b = 2 - 1 = 1$$

$$c = C_0 - (a + b) = 0 - 1 = -1$$

$$C_1 - c = a\varphi - \frac{1-a}{\varphi} = 1 + 1 = 2$$

$$a\left(\varphi^2 + 1\right) = 2\varphi + 1$$

$$a = \frac{2\varphi + 1}{\varphi + 2} = \frac{\varphi^3}{\varphi + 2}$$

$$b = 1 - a = \frac{1-\varphi}{\varphi+2} = -\frac{1}{\varphi(\varphi+2)}$$

$$C_h = \frac{1}{\varphi+2}\left(\varphi^{h+3} + \left(-\frac{1}{\varphi}\right)^{h+1}\right) - 1$$

$$\frac{1}{\varphi+2} = \frac{2}{5+\sqrt{5}} = \frac{1}{\sqrt{5}\varphi}$$

$$C_h = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{h+2} - \left(\frac{1-\sqrt{5}}{2}\right)^{h+2}\right) - 1$$

$$B_h = 2^{\frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^{h+2} - \left(\frac{1-\sqrt{5}}{2}\right)^{h+2}\right)-1}$$

□

VII Splay tree

i Splay tree

A splay tree is a self-adjusting binary search tree (BST) that keeps no info of height and in which every access operation moves the accessed node to the root through an operation called splaying, invented by Daniel Dominic Sleator and Robert Endre Tarjan. The idea is locality, that is, if an item is accessed, it is likely to be accessed again soon. The worst-case searching, insertion, and deletion time complexities are as with a normal BST, while the amortized time complexities are $O(\log n)$.

ii (Bottom-up) splay

When a node x is accessed, a (bottom-up) splay operation consisting of a sequence of splay steps is performed on x to move it to the root, called splaying x. Let the parent of x be p and the parent of p be g. There are three types of splay steps, each of which has two symmetric variants:

- Zig step / simple rotation: This is done if x is a child of the root. We perform a rotation on the edge between root and x. This only happens as the last step and only happens when x originally has odd depth.
- Zig-zig step: This is done either when x is a left child of p and p is a left child of g, or when x is a right child of p and p is a right child of g. We perform a rotation on the edge between g and p and then a rotation on the edge between p and x.
- Zig-zag step: This is done either when x is a left child of p and p is a right child of g, or when x is a right child of p and p is a left child of g. We perform a rotation on the edge between p and x, and then a rotation on the edge between g and x, now at the original position of p.

A zig step is a 1-level splay step while a zig-zig or zig-zag step is a 2-level splay step.

iii Searching and finding minimum/maximum

Searching and finding minimum/maximum in splay tree involves first doing the operation as with a normal BST and then splay the found node if found.

iv Join

Given two BSTs S and T such that all elements of S are smaller than all elements of T, the following steps can be used to join them to a single BST: either

1. Splay the largest item in S. Now this item is in the root r of S and has no right child.
2. Set the right child of the new root r of S to the root of T.

or

1. Splay the smallest item in T. Now this item is in the root r of T and has no left child.
2. Set the left child of the new root r of T to the root of S.

v Split

Given a BST and a value x, splitting the BST at x returns two new BSTs, one containing all elements less than or equal to x, L, and the other containing all elements greater than x, R. This can be done with:

1. Search for x in the tree. If we found it, let y be that node; otherwise let y be the node we stop at.
2. Splay y.
3. Split its right subtree, which is R, from the rest of the tree, which is L.

vi Insertion

To insert a value x into a splay tree, there are two ways: either

1. Insert x as with a normal BST.
2. Splay x.

or

1. Split at x. However, if x is found in the tree, stop after splaying x.
2. Create a new tree with x as root and the two split trees as its subtrees.

vii Deletion

To delete a node x from a splay tree: either

1. Search for x.
2. Splay x.
3. Delete x and join the two subtrees of x.

or

1. Delete x as with a normal BST.
2. Splay the original parent of the deleted x.

viii Amortized analysis

We define size of a node $s(x)$ as the number of nodes in the subtree whose root is node x, rank of a node as

$$r(x) = \log_2 s(x)$$

and potential function of a tree $P(T)$ is

$$P(T) = \sum_{x \in T} r(x)$$

The actual cost C_i of an operation i is defined as the number of rotations in the operation. Let the state before operation i be denoted by the original functions and the state after operation i be denoted with a prime symbol. The amortized cost A_i of an operation i with actual cost C_i that transforms the tree T_{i-1} to T_i is defined as

$$A_i = C_i + \Delta P = C_i + P(T_i) - P(T_{i-1})$$

- Zig step:

$$C_i = 1$$

$$r'(p) \leq r(p)$$

$$\Delta P = r'(x) + r'(p) - r(x) - r(p) \leq r'(x) - r(x)$$

$$A_i \leq 1 + r'(x) - r(x)$$

- Zig-zig step:

$$C_i = 2$$

$$r'(p) \leq r(g)$$

$$r(x) \leq r(p)$$

$$s(x) + s'(g) = s'(x) - 2 \leq s'(x)$$

The log function is concave, i.e.,

$$\frac{\log_2 a + \log_2 b}{2} \leq \log_2\left(\frac{a+b}{2}\right)$$

i.e.,

$$log_2 a + \log_2 b \leq 2\log_2(a+b) - 2$$

Hence

$$r(x) + r'(g) = 2r'(x) - 2$$

$$r'(g) \leq 2r'(x) - r(x) - 2$$

$$\Delta P = r'(x) + r'(p) + r'(g) - r(x) - r(p) - r(g) \leq r'(x) + r'(g) - 2r(x) \leq 3(r'(x) - r(x)) - 2$$

$$A_i \leq 3(r'(x) - r(x))$$

- Zig-zag step:

$$C_i = 2$$

$$s'(p) + s'(g) = s'(x) - 1 \leq s'(x)$$

The log function is concave, i.e.,

$$\frac{\log_2 a + \log_2 b}{2} \leq \log_2\left(\frac{a+b}{2}\right)$$

i.e.,

$$log_2 a + \log_2 b \leq 2\log_2(a+b) - 2$$

Hence

$$r'(p) + r'(g) \leq 2r'(x) - 2$$

$$r(x) \leq r(p) \leq r(g)$$

$$\Delta P = r'(x) + r'(p) + r'(g) - r(x) - r(p) - r(g) \leq 3(r'(x) - r(x)) - 2$$

$$A_i \leq 3(r'(x) - r(x))$$

When splaying a node x, zero or more 2-level splay steps followed by zero or one 1-level splay step is conducted. Let $r_2(x)$ be the rank of x just after the last 2-level splay step and $r_f(x)$ be the final rank of x. The amortized cost A of splaying x is

$$\begin{aligned} A &\leq 1 + r_f(x) - r_2(x) + 3(r_2(x) - r(x)) \\ &\leq 1 + 3(r_f(x) - r_2(x)) + 3(r_2(x) - r(x)) \\ &= 1 + 3(r_f(x) - r(x)) \\ &\leq 1 + 3(\log_2 n - r(x)) \\ &= O(\log n) \end{aligned}$$

For a sequence of m accesses in a splay tree T, we can bound the total actual cost with

$$\sum_{i=1}^{m} C_i = P(T_0) - P(T_m) + \sum_{i=1}^{m} A_i$$

For every node x,

$$1 \leq s(x) \leq n$$

$$0 \leq r(x) \leq \log_2 n$$

Thus the change of rank $\Delta r(x)$ of a node x is

$$\Delta r(x) \leq \log_2 n$$

Thus the change of potential function is

$$P(T_0) - P(T_m) \leq \sum_{x \in T} \log_2 n = O\left(n \log n\right)$$

ix Weighted analysis

The above analysis can be generalized by assigning each node x a weight $w(x) \geq 1$ and defining $s(x)$ as the sum of weights of nodes in the subtree whose root is node x, called weighted analysis. Note that the previous analysis corresponds to the special case $w(x) = 1$. The sum of all weights is denoted W. The same analysis gives:

$$A \leq 1 + 3(\log_2 n - r(x)) = O(\log\left(\frac{W}{w(x)}\right) + 1)$$

For a sequence of m accesses in a splay tree T, we can bound the total actual cost with

$$\sum_{i=1}^{m} C_i = P(T_0) - P(T_m) + \sum_{i=1}^{m} A_i$$

For every node x,

$$w(x) \leq s(x) \leq W$$

$$\log_2(w(x)) \leq r(x) \leq \log_2 W$$

Thus the change of rank $\Delta r(x)$ of a node x is

$$\Delta r(x) \leq \log_2\left(\frac{W}{w(x)}\right)$$

Thus the change of potential function is

$$P(T_0) - P(T_m) \leq \sum_{x \in T} \log_2\left(\frac{W}{w(x)}\right) = O\left(\sum_{x \in T} \log\left(\frac{W}{w(x)}\right)\right)$$

x Balance theorem

The cost of performing a sequence of m accesses in a splay tree containing n nodes is

$$O(m \log n + n \log n)$$

Proof. Take a constant weight, e.g.,

$$w(x) = 1$$

Then

$$W = n$$

$$A_i = O(\log n)$$

$$P(T_0) - P(T_m) = O(n \log n)$$

□

This theorem implies that splay trees perform at least as well as AVL trees on sequences of at least n accesses.

xi Static optimality theorem

The cost of performing a sequence of m accesses in a splay tree T containing n nodes, with each node accessed at least once and q_x being the number of times node x is accessed, is

$$O\left(m + \sum_{x \in T} q_x \log\left(\frac{m}{q_x}\right)\right)$$

Proof. Let

$$w(x) = q_x$$

Then

$$W = m$$

$$A_i = O\left(\log\left(\frac{m}{q_x}\right) + 1\right)$$

$$\sum_{i=1}^{m} A_i = O\left(\sum_{x \in T} q_x \log\left(\frac{m}{q_x}\right) + m\right)$$

$$q_x \geq 1$$

$$P(T_0) - P(T_m) = O\left(\sum_{x \in T} \log\left(\frac{m}{q_x}\right)\right) = O\left(\sum_{x \in T} q_x \log\left(\frac{m}{q_x}\right)\right)$$

□

xii Working set theorem

The cost of performing a sequence S of m accesses in a splay tree T containing n nodes, with $t(x)$ being the number of distinct nodes accessed before the previous time node x was accessed for each $x \in S$, is

$$O\left(m + n \log n + \sum_{x \in S} \log(t(x) + 1)\right)$$

Proof. Let

$$w(x) = (t(x) + 1)^{-2}$$

Then

$$W = O(1)$$

$$A_i = O\left(\log(t(x) + 1) + 1\right)$$

$$\sum_{i=1}^{m} A_i = O\left(\sum_{x \in S} \log(t(x) + 1) + m\right)$$

$$P(T_0) - P(T_m) = O\left(n \log n\right)$$

□

xiii Locality property or strong balance theorem

The cost of performing a sequence of m accesses in a splay tree containing n nodes, with k distinct nodes accessed, is

$$O(m + m \log k + n \log n)$$

Proof. Use working set theorem.

$$t(x) = O(k)$$

$$O\left(m + n \log n + \sum_{x \in S} \log(t(x) + 1)\right) = O(m + n \log n + m \log k)$$

□

xiv Top-down splay

An alternative implementation of splay that is done together with traversal from root to the node x to be accessed. Top-down splay also consists of a sequence of splay steps. We start at the root and maintain two trees, left tree L of nodes smaller than x and right tree R of nodes larger than x. The original tree is called middle or center tree. In each step, the current node in the middle tree is denoted m, which is root of the original tree, the current node in the left tree is denoted l, and the current node of the right tree is denoted r. We first define linking, which has two symmetric variants:

- Right linking: If R is nonempty, connect m as the left child of r and then assign r as it; if R is empty, assign m as both the root of R and r; and then detach the left child of m and assign it as the new m.
- Left linking: If L is nonempty, connect m as the right child of l and then assign l as it; if L is empty, assign m as both the root of L and l; and then detach the right child of m and assign it as the new m.

There are three types of splay steps, each of which has two symmetric variants:

- LL case zig-zig step: This is done when x is in left(left)(m), i.e., $x < m.\text{left}$. We perform a right rotation on the edge between m and its left child, and assign its original left child, now parent, as the new m, and then,
  - if m has left child, we perform a right linking.
  - if m has no left child, we stop the traversal and go to the last step.
- RR case zig-zig step: This is done when x is in right(right)(m), i.e., $x > m.\text{right}$. We perform a left rotation on the edge between m and its right child, and assign its original right child, now parent, as the new m, and then,
  - if m has right child, we perform a left linking.

- if m has no right child, we stop the traversal and go to the last step.

- LR case zig-zag step: This is done when x is in right(left)(m), i.e., $m > x \geq m.$left. We perform a right linking.
- RL case zig-zag step: This is done when x is in left(right)(m), i.e., $m < x \leq m.$right. We perform a left linking.
- $x = m$, $x < m$ and m has no left child, or $x > m$ and m has no right child: We stop the traversal and go to the last step.
- Last step:
  - If L is nonempty, connect left(m) as right(l), and connect the root of L as left(m);
  - if R is nonempty, connect right(m) as left(r), and connect the root of R as right(m).

Top-down splay has the same worst-case and amortized searching, insertion, and deletion time complexities as bottom-up splay and $O(1)$ extra space.

VIII An implementation of binary tree, BST, ASL tree, and splay tree in C++

```cpp
#pragma once
#include <algorithm>
#include <cctype>
#include <complex>
#include <concepts>
#include <cstddef>
#include <functional>
#include <istream>
#include <ostream>
#include <sstream>
#include <stdexcept>
#include <string>
#include <string_view>
#include <utility>
#include <vector>

template <class T, class Node> class BaseBinaryTreeNode {
public:
  using value_type = T;
  using node_type = Node;

  T value;
  Node *parent;
  Node *left = nullptr;
  Node *right = nullptr;

  BaseBinaryTreeNode(const T &_value = T(), Node *_parent = nullptr)
      : value(_value), parent(_parent) {}

  virtual ~BaseBinaryTreeNode() = default;

```

```
  virtual void _clone_helper(Node *) const {}
};

template <class T>
class BinaryTreeNode : public BaseBinaryTreeNode<T, BinaryTreeNode<T>> {
public:
  using Base = BaseBinaryTreeNode<T, BinaryTreeNode<T>>;
  using Base::Base;
  using value_type = T;
  using node_type = Base::node_type;
};

template <class T, class Node = BinaryTreeNode<T>> class BinaryTreeStatic {
public:
  using node_type = Node;

  static std::vector<T> preorder(Node *node) {
    if (node == nullptr)
      return std::vector<T>();
    std::vector<T> result;
    std::vector<Node *> stack;
    stack.push_back(node);
    while (!stack.empty()) {
      Node *cur = stack.back();
      stack.pop_back();
      result.push_back(cur->value);
      if (cur->right != nullptr)
        stack.push_back(cur->right);
      if (cur->left != nullptr)
        stack.push_back(cur->left);
    }
    return result;
  }

  static std::vector<Node *> preorder_node(Node *node) {
    if (node == nullptr)
      return std::vector<Node *>();
    std::vector<Node *> result, stack;
    stack.push_back(node);
    while (!stack.empty()) {
      Node *cur = stack.back();
      stack.pop_back();
      result.push_back(cur);
      if (cur->right != nullptr)
        stack.push_back(cur->right);
      if (cur->left != nullptr)
        stack.push_back(cur->left);
    }
    return result;
  }

  static std::vector<T> inorder(Node *node) {
    if (node == nullptr)
      return std::vector<T>();
    std::vector<T> result;
    std::vector<Node *> stack;
    while (!stack.empty() || node != nullptr) {
      while (node != nullptr) {
        stack.push_back(node);
        node = node->left;
      }
      node = stack.back();
```

```
      stack.pop_back();
      result.push_back(node->value);
      node = node->right;
    }
    return result;
  }

  static std::vector<Node *> inorder_node(Node *node) {
    if (node == nullptr)
      return std::vector<Node *>();
    std::vector<Node *> result, stack;
    while (!stack.empty() || node != nullptr) {
      while (node != nullptr) {
        stack.push_back(node);
        node = node->left;
      }
      node = stack.back();
      stack.pop_back();
      result.push_back(node);
      node = node->right;
    }
    return result;
  }

  static std::vector<T> postorder(Node *node) {
    if (node == nullptr)
      return std::vector<T>();
    std::vector<T> result;
    std::vector<Node *> stack;
    Node *last_visited = nullptr;
    while (!stack.empty() || node != nullptr) {
      if (node != nullptr) {
        stack.push_back(node);
        node = node->left;
      } else {
        Node *right = stack.back()->right;
        if (right != nullptr && last_visited != right)
          node = right;
        else {
          result.push_back(stack.back()->value);
          last_visited = stack.back();
          stack.pop_back();
        }
      }
    }
    return result;
  }

  static std::vector<Node *> postorder_node(Node *node) {
    if (node == nullptr)
      return std::vector<Node *>();
    std::vector<Node *> result, stack;
    Node *last_visited = nullptr;
    while (!stack.empty() || node != nullptr) {
      if (node != nullptr) {
        stack.push_back(node);
        node = node->left;
      } else {
        Node *right = stack.back()->right;
        if (right != nullptr && last_visited != right)
          node = right;
        else {
```

```
          result.push_back(stack.back());
          last_visited = stack.back();
          stack.pop_back();
        }
      }
    }
    return result;
  }

  // 1: left child
  // 0: right child
  // -1: node's parent is nullptr
  // -2: node is nullptr
  static signed char is_left_child(Node *node) noexcept {
    if (node == nullptr)
      return -2;
    if (node->parent == nullptr)
      return -1;
    return node->parent->left == node;
  }

  static Node *leftmost_child(Node *node) {
    if (node == nullptr)
      return nullptr;
    while (node->left != nullptr)
      node = node->left;
    return node;
  }

  static Node *rightmost_child(Node *node) {
    if (node == nullptr)
      return nullptr;
    while (node->right != nullptr)
      node = node->right;
    return node;
  }

  // LaTeX forest representation: [value[left][right]]
  // A missing child whose sibling exists is emitted as [,phantom]
  static void latex_forest(Node *node, std::ostream *oss, bool outer = true) {
    if (node == nullptr || oss == nullptr)
      return;
    if (outer)
      *oss << "\\begin{center}\n\\begin{forest}\n";
    *oss << "[" << node->value;
    bool has_left = (node->left != nullptr);
    bool has_right = (node->right != nullptr);
    if (has_left)
      latex_forest(node->left, oss, false);
    else if (has_right)
      *oss << "[,phantom]";
    if (has_right)
      latex_forest(node->right, oss, false);
    else if (has_left)
      *oss << "[,phantom]";
    *oss << "]";
    if (outer)
      *oss << "\n\\end{forest}\n\\end{center}\n";
  }

  // O(n)
  // Deep-clones the subtree rooted at `node` with given parent of new root used
```

```
  // Return the new root
  static Node *clone_subtree(const Node *node, Node *parent = nullptr) {
    if (node == nullptr)
      return nullptr;
    Node *copy = new Node(node->value, parent);
    node->_clone_helper(copy);
    copy->left = clone_subtree(node->left, copy);
    copy->right = clone_subtree(node->right, copy);
    return copy;
  }
};

// Take LaTeX forest representation: [value[left][right]]
// Output root node
template <class T, class Node = BinaryTreeNode<T>>
  requires std::default_initializable<T> &&
           requires(std::istream &is, T &value) {
             { is >> value } -> std::same_as<std::istream &>;
           }
Node *from_latex_forest(std::string s) {
  std::erase_if(s, [](unsigned char c) { return std::isspace(c); });
  if (s.empty())
    return nullptr;
  std::size_t pos = 0;
  std::vector<Node *> owned;
  auto consume = [&](char expected) {
    if (pos >= s.size() || s[pos] != expected)
      throw std::invalid_argument(std::string("from_latex_forest: expected '") +
                                  expected + "'");
    ++pos;
  };
  auto parse_node = [&](auto &&self) -> Node * {
    consume('[');
    if (pos + 9 <= s.size() && s[pos] == ',' &&
        s.compare(pos + 1, 7, "phantom") == 0 && s[pos + 8] == ']') {
      pos += 9;
      return nullptr;
    }
    std::size_t start = pos;
    while (pos < s.size() && s[pos] != '[' && s[pos] != ']')
      ++pos;
    std::string value_str = s.substr(start, pos - start);
    std::istringstream vs(value_str);
    T value{};
    if (!(vs >> value))
      throw std::invalid_argument("from_latex_forest: bad value '" + value_str +
                                  "'");
    Node *node = new Node(value);
    owned.push_back(node);
    if (pos < s.size() && s[pos] == '[') {
      node->left = self(self);
      if (node->left)
        node->left->parent = node;
      node->right = self(self);
      if (node->right)
        node->right->parent = node;
    }
    consume(']');
    return node;
  };
  Node *root = nullptr;
  try {
```

```
    root = parse_node(parse_node);
    if (pos != s.size())
      throw std::invalid_argument("from_latex_forest: trailing characters");
  } catch (...) {
    for (Node *n : owned)
      delete n;
    throw;
  }
  return root;
}

template <class T, class Node = BinaryTreeNode<T>>
class BinaryTree : public BinaryTreeStatic<T, Node> {
public:
  using node_type = Node;
  using Base = BinaryTreeStatic<T, Node>;
  using Base::clone_subtree;
  using Base::inorder;
  using Base::inorder_node;
  using Base::is_left_child;
  using Base::latex_forest;
  using Base::leftmost_child;
  using Base::postorder;
  using Base::postorder_node;
  using Base::preorder;
  using Base::preorder_node;
  using Base::rightmost_child;

  mutable Node *root; // Allow root->parent != nullptr
  std::ostream *oss;  // latex_forest logger

  BinaryTree(Node *_root = nullptr, std::ostream *_oss = nullptr)
      : root(_root), oss(_oss) {}

  BinaryTree(const BinaryTree &other)
      : root(clone_subtree(other.root,
                           other.root ? other.root->parent : nullptr)),
        oss(other.oss) {}

  BinaryTree(BinaryTree &&other) noexcept
      : root(std::exchange(other.root, nullptr)),
        oss(std::exchange(other.oss, nullptr)) {}

  void set_logger(std::ostream *_oss) { oss = _oss; }

  bool is_empty() const { return root == nullptr; }

  // O(n)
  std::vector<T> preorder() const { return preorder(root); }
  std::vector<Node *> preorder_node() const { return preorder_node(root); }
  std::vector<T> inorder() const { return inorder(root); }
  std::vector<Node *> inorder_node() const { return inorder_node(root); }
  std::vector<T> postorder() const { return postorder(root); }
  std::vector<Node *> postorder_node() const { return postorder_node(root); }

  // O(h)
  Node *leftmost_child() const { return leftmost_child(root); }
  Node *rightmost_child() const { return rightmost_child(root); }

  virtual void log(std::string_view msg) const {
    if (oss != nullptr)
      *oss << msg << '\n';
```

```
342   }
343
344   virtual void log_forest(Node *node) const {
345     if (oss != nullptr)
346       latex_forest(node, oss);
347   }
348
349   // O(n)
350   virtual void log_forest() const { return log_forest(root); }
351
352 protected:
353   // O(1)
354   virtual Node *replace(Node *node, Node *replacement,
355                         bool deleting = false) const {
356     switch (is_left_child(node)) {
357     case -2:
358       throw std::invalid_argument("Node is nullptr");
359       break;
360     case 1:
361       node->parent->left = replacement;
362       break;
363     case 0:
364       node->parent->right = replacement;
365       break;
366     }
367     if (root == node)
368       root = replacement;
369     if (replacement != nullptr)
370       replacement->parent = node->parent;
371     Node *parent = node->parent;
372     if (deleting)
373       delete node;
374     return parent;
375   }
376
377   void recursive_clear(Node *node) noexcept {
378     if (node == nullptr)
379       return;
380     recursive_clear(node->left);
381     recursive_clear(node->right);
382     delete node;
383   }
384
385 public:
386   // recursive:
387   // >0: stack
388   // 0: try stack catch recursive
389   // <0: recursive
390   void clear(Node *node, signed char recursive = 0) noexcept {
391     if (node == nullptr)
392       return;
393     if (root == node)
394       root = nullptr;
395     switch (is_left_child(node)) {
396     case 1:
397       node->parent->left = nullptr;
398       break;
399     case 0:
400       node->parent->right = nullptr;
401       break;
402     }
403     auto stack_clear = [](Node *node) {
```

```
404       std::vector<Node *> stack = preorder_node(node);
405       for (auto it = stack.rbegin(); it != stack.rend(); ++it)
406         delete *it;
407     };
408     if (recursive > 0)
409       stack_clear(node);
410     else if (recursive == 0) {
411       try {
412         stack_clear(node);
413       } catch (...) {
414         recursive_clear(node);
415       }
416     } else
417       recursive_clear(node);
418   }
419
420   void clear() noexcept { clear(root); }
421
422   virtual ~BinaryTree() { clear(); }
423
424   BinaryTree &operator=(const BinaryTree &other) {
425     if (this == &other)
426       return *this;
427     clear();
428     root = clone_subtree(other.root, other.root ? other.root->parent : nullptr);
429     oss = other.oss;
430     return *this;
431   }
432
433   BinaryTree &operator=(BinaryTree &&other) noexcept {
434     if (this == &other)
435       return *this;
436     clear();
437     root = std::exchange(other.root, nullptr);
438     oss = std::exchange(other.oss, nullptr);
439     return *this;
440   }
441
442   // O(1)
443   virtual Node *rotate_right(Node *node) const {
444     if (node == nullptr || node->left == nullptr)
445       throw std::invalid_argument("Node is nullptr");
446     Node *child = node->left;
447     replace(node, child);
448     node->left = child->right;
449     if (child->right != nullptr)
450       child->right->parent = node;
451     child->right = node;
452     node->parent = child;
453     log("Rotate right:");
454     log_forest();
455     return child;
456   }
457
458   // O(1)
459   virtual Node *rotate_left(Node *node) const {
460     if (node == nullptr || node->right == nullptr)
461       throw std::invalid_argument("Node is nullptr");
462     Node *child = node->right;
463     replace(node, child);
464     node->right = child->left;
465     if (child->left != nullptr)
```

```
      child->left->parent = node;
    child->left = node;
    node->parent = child;
    log("Rotate left:");
    log_forest();
    return child;
  }

  // O(1)
  virtual Node *rotate_parent(Node *child) const {
    if (child == nullptr || child->parent == nullptr)
      throw std::invalid_argument("Node is nullptr");
    if (is_left_child(child) == 1)
      return rotate_right(child->parent);
    return rotate_left(child->parent);
  }
};

template <class T, class Node = BinaryTreeNode<T>, class Compare = std::less<T>>
  requires std::strict_weak_order<const Compare &, const T &, const T &>
class BST : public BinaryTree<T, Node> {
public:
  using node_type = Node;
  using Base = BinaryTree<T, Node>;
  using Base::clear;
  using Base::inorder;
  using Base::inorder_node;
  using Base::leftmost_child;
  using Base::log;
  using Base::log_forest;
  using Base::postorder;
  using Base::postorder_node;
  using Base::preorder;
  using Base::preorder_node;
  using Base::rightmost_child;
  using Base::root;
  using Base::rotate_left;
  using Base::rotate_right;

  [[no_unique_address]] Compare comp;

protected:
  using Base::replace;

public:
  BST(Node *_root = nullptr, std::ostream *_oss = nullptr,
      const Compare &_comp = Compare())
      : Base(_root, _oss), comp(_comp) {}
  explicit BST(const Compare &_comp) : BST(nullptr, nullptr, _comp) {}

  BST(const BST &other) : Base(other), comp(other.comp) {}
  BST(BST &&other) : Base(other), comp(other.comp) {}

  BST &operator=(const BST &other) {
    if (this != &other) {
      Base::operator=(other);
      comp = other.comp;
    }
    return *this;
  }

  BST &operator=(BST &&other) {
```

```
528     if (this != &other) {
529       Base::operator=(other);
530       comp = std::exchange(other.comp, nullptr);
531     }
532     return *this;
533   }
534
535   // O(h)
536   virtual Node *find_min() const { return leftmost_child(root); }
537   virtual Node *find_max() const { return rightmost_child(root); }
538
539   // Return node, found
540   // O(h)
541   virtual std::pair<Node *, bool> search(const T &x) const {
542     if (root == nullptr)
543       return {nullptr, false};
544     Node *node = root;
545     while (true) {
546       if (comp(x, node->value)) {
547         if (node->left == nullptr)
548           return {node, false};
549         node = node->left;
550       } else if (comp(node->value, x)) {
551         if (node->right == nullptr)
552           return {node, false};
553         node = node->right;
554       } else
555         return {node, true};
556     }
557   }
558
559   // O(h)
560   Node *inorder_successor(Node *node) const {
561     if (node == nullptr)
562       throw std::invalid_argument("Node is nullptr");
563     if (node->right != nullptr)
564       return leftmost_child(node->right);
565     if (node == root)
566       return nullptr;
567     Node *p = node->parent;
568     while (p != nullptr && node == p->right) {
569       if (p == root)
570         return nullptr;
571       node = p;
572       p = p->parent;
573     }
574     return p;
575   }
576
577   // O(h)
578   Node *inorder_predecessor(Node *node) const {
579     if (node == nullptr)
580       throw std::invalid_argument("Node is nullptr");
581     if (node->left != nullptr)
582       return rightmost_child(node->left);
583     if (node == root)
584       return nullptr;
585     Node *p = node->parent;
586     while (p != nullptr && node == p->left) {
587       if (p == root)
588         return nullptr;
589       node = p;
```

```
590       p = p->parent;
591     }
592     return p;
593   }
594
595   // O(h)
596   virtual Node *ordered_after(const T &x) const {
597     Node *result = nullptr;
598     Node *node = root;
599     while (node != nullptr) {
600       if (comp(x, node->value)) {
601         result = node;
602         node = node->left;
603       } else
604         node = node->right;
605     }
606     return result;
607   }
608
609   // O(h)
610   virtual Node *ordered_not_before(const T &x) const {
611     Node *result = nullptr;
612     Node *node = root;
613     while (node != nullptr) {
614       if (!comp(node->value, x)) {
615         result = node;
616         node = node->left;
617       } else
618         node = node->right;
619     }
620     return result;
621   }
622
623   // O(h)
624   virtual Node *ordered_before(const T &x) const {
625     Node *result = nullptr;
626     Node *node = root;
627     while (node != nullptr) {
628       if (comp(node->value, x)) {
629         result = node;
630         node = node->right;
631       } else
632         node = node->left;
633     }
634     return result;
635   }
636
637   // O(h)
638   virtual Node *ordered_not_after(const T &x) const {
639     Node *result = nullptr;
640     Node *node = root;
641     while (node != nullptr) {
642       if (!comp(x, node->value)) {
643         result = node;
644         node = node->right;
645       } else
646         node = node->left;
647     }
648     return result;
649   }
650
651   // Return node, inserted
```

```
652   // O(h)
653   virtual std::pair<Node *, bool> insert(const T &x) {
654     if (root == nullptr) {
655       root = new Node(x);
656       log("Insert:");
657       log_forest();
658       return {root, true};
659     }
660     Node *node = root;
661     while (true) {
662       if (comp(x, node->value)) {
663         if (node->left == nullptr) {
664           node->left = new Node(x, node);
665           log("Insert:");
666           log_forest();
667           return {node->left, true};
668         }
669         node = node->left;
670       } else if (comp(node->value, x)) {
671         if (node->right == nullptr) {
672           node->right = new Node(x, node);
673           log("Insert:");
674           log_forest();
675           return {node->right, true};
676         }
677         node = node->right;
678       } else
679         return {node, false};
680     }
681   }
682
683   // Return rebalance_start
684   // O(h)
685   virtual Node *remove_node(Node *node) {
686     if (node == nullptr)
687       throw std::invalid_argument("Node is nullptr");
688     if (node->left == nullptr) {
689       log("Delete:");
690       Node *result = replace(node, node->right, true);
691       log_forest();
692       return result;
693     }
694     if (node->right == nullptr) {
695       log("Delete:");
696       Node *result = replace(node, node->left, true);
697       log_forest();
698       return result;
699     }
700     Node *succ = inorder_successor(node);
701     Node *rebalance_start;
702     if (succ->parent != node) {
703       rebalance_start = succ->parent;
704       replace(succ, succ->right);
705       succ->right = node->right;
706       node->right->parent = succ;
707     } else
708       rebalance_start = succ;
709     replace(node, succ);
710     succ->left = node->left;
711     succ->left->parent = succ;
712     delete node;
713     log("Delete:");
```

```
    log_forest();
    return rebalance_start;
  }

  // Return rebalance_start, removed
  // O(h)
  virtual std::pair<Node *, bool> remove(const T &x) {
    auto [node, found] = search(x);
    if (!found)
      return {node, false};
    return {remove_node(node), true};
  }
};

template <class T>
class AVLTreeNode : public BaseBinaryTreeNode<T, AVLTreeNode<T>> {
public:
  using Base = BaseBinaryTreeNode<T, AVLTreeNode<T>>;
  using Base::Base;
  using Base::left;
  using Base::right;
  using value_type = T;
  using node_type = Base::node_type;

  mutable int height = 1; // leaf 1

  void _clone_helper(AVLTreeNode *other) const override {
    other->height = height;
  }

  static int get_height(AVLTreeNode *node) noexcept {
    return node == nullptr ? 0 : node->height;
  }

  static int bf(AVLTreeNode *node) {
    return get_height(node->left) - get_height(node->right);
  }

  void update_height() const {
    height = 1 + std::max(get_height(left), get_height(right));
  }
};

template <class T, class Node = AVLTreeNode<T>, class Compare = std::less<T>>
class AVLTree : public BST<T, Node, Compare> {
public:
  using node_type = Node;
  using Base = BST<T, Node, Compare>;
  using Base::Base;
  using Base::log;
  using Base::log_forest;
  using Base::root;

  AVLTree(const AVLTree &) = default;
  AVLTree(AVLTree &&) = default;
  AVLTree &operator=(const AVLTree &) = default;
  AVLTree &operator=(AVLTree &&) = default;

  Node *rotate_left(Node *node) const override {
    Node *child = Base::rotate_left(node);
    node->update_height();
    child->update_height();
```

```
776     return child;
777   }
778
779   Node *rotate_right(Node *node) const override {
780     Node *child = Base::rotate_right(node);
781     node->update_height();
782     child->update_height();
783     return child;
784   }
785
786   // Rebalance until rebalanced once or parent == nullptr
787   // up to outer tree if root->parent != nullptr
788   virtual Node *rebalance(Node *node) const {
789     node->update_height();
790     int bf = Node::bf(node);
791     if (bf > 1) {
792       Node *child = node->left;
793       if (Node::bf(child) < 0) {
794         log("LR case left-then-right double rotation:\n");
795         rotate_left(child);
796       } else
797         log("LL case right single rotation:\n");
798       return rotate_right(node);
799     }
800     if (bf < -1) {
801       Node *child = node->right;
802       if (Node::bf(child) > 0) {
803         log("RL case right-then-left double rotation:\n");
804         rotate_right(child);
805       } else
806         log("RR case left single rotation:\n");
807       return rotate_left(node);
808     }
809     if (node->parent == nullptr)
810       return node;
811     return rebalance(node->parent);
812   }
813
814   std::pair<Node *, bool> insert(const T &x) override {
815     auto [node, inserted] = Base::insert(x);
816     if (inserted && node->parent != nullptr)
817       rebalance(node->parent);
818     return {node, inserted};
819   }
820
821   Node *remove_node(Node *node) override {
822     Node *rebalance_start = Base::remove_node(node);
823     Node *cur = rebalance_start;
824     while (cur != nullptr) {
825       cur = rebalance(cur)->parent;
826     }
827     return rebalance_start;
828   }
829 };
830
831 template <class T, class Node = BinaryTreeNode<T>, class Compare = std::less<T>>
832 class SplayTree : public BST<T, Node, Compare> {
833 public:
834   using node_type = Node;
835   using Base = BST<T, Node, Compare>;
836   using Base::Base;
837   using Base::comp;
```

```
838   using Base::is_left_child;
839   using Base::log;
840   using Base::log_forest;
841   using Base::oss;
842   using Base::root;
843   using Base::rotate_left;
844   using Base::rotate_parent;
845   using Base::rotate_right;
846
847 protected:
848   using Base::replace;
849
850 public:
851   SplayTree(const SplayTree &) = default;
852   SplayTree(SplayTree &&) = default;
853   SplayTree &operator=(const SplayTree &) = default;
854   SplayTree &operator=(SplayTree &&) = default;
855
856   // O(h) worst case, O(log n) amortized
857   virtual Node *splay(Node *node) const {
858     if (node == nullptr)
859       return nullptr;
860     while (node != root) {
861       Node *parent = node->parent;
862       Node *grandparent = parent->parent;
863       if (parent == root) {
864         log("Zig step:\n");
865         rotate_parent(node);
866       } else if (parent->left == node && grandparent->left == parent) {
867         log("Zig-zig step:\n");
868         rotate_right(grandparent);
869         rotate_right(parent);
870       } else if (parent->right == node && grandparent->right == parent) {
871         log("Zig-zig step:\n");
872         rotate_left(grandparent);
873         rotate_left(parent);
874       } else {
875         log("Zig-zag step:\n");
876         rotate_parent(node);
877         rotate_parent(node);
878       }
879     }
880     return node;
881   }
882
883   Node *find_min() const override { return splay(Base::find_min()); }
884   Node *find_max() const override { return splay(Base::find_max()); }
885
886   Node *ordered_after(const T &x) const override {
887     return splay(Base::ordered_after(x));
888   }
889
890   Node *ordered_not_before(const T &x) const override {
891     return splay(Base::ordered_not_before(x));
892   }
893
894   Node *ordered_not_after(const T &x) const override {
895     return splay(Base::ordered_not_after(x));
896   }
897
898   Node *ordered_before(const T &x) const override {
899     return splay(Base::ordered_before(x));
```

```
  }

  std::pair<Node *, bool> search(const T &x) const override {
    auto [result, second] = Base::search(x);
    return {splay(result), second};
  }

  std::pair<Node *, bool> insert(const T &x) override {
    auto [result, second] = Base::insert(x);
    return {splay(result), second};
  }

  virtual void join_right(Node *other) {
    if (other == nullptr)
      return;
    log("Join right:\n");
    log("To join:");
    log_forest(other);
    if (root == nullptr) {
      root = other;
      other->parent = nullptr;
    } else {
      find_max();
      root->right = other;
      other->parent = root;
    }
    log("Join:");
    log_forest();
  }

  virtual void join_left(Node *other) {
    if (other == nullptr)
      return;
    log("Join left:\n");
    log("To join:");
    log_forest(other);
    if (root == nullptr) {
      root = other;
      other->parent = nullptr;
    } else {
      find_min();
      root->left = other;
      other->parent = root;
    }
    log("Join:");
    log_forest();
  }

  // Not return rebalance_start but always return nullptr if node != nullptr
  Node *remove_node(Node *node) override {
    log("Delete:\n");
    if (node == nullptr)
      throw std::invalid_argument("Node is nullptr");
    splay(node);
    Node *right = root->right;
    if (root->left != nullptr) {
      replace(root, root->left, true);
      if (right != nullptr)
        right->parent = nullptr;
      join_right(right);
    } else {
      replace(root, right, true);
```

```
962       log_forest();
963     }
964     return nullptr;
965   }
966
967 protected:
968   virtual SplayTree split_right(Node *other_root) {
969     if (other_root == nullptr)
970       return SplayTree(nullptr, oss, comp);
971     replace(other_root, other_root->left);
972     other_root->parent = nullptr;
973     other_root->left = nullptr;
974     SplayTree result = SplayTree(other_root, oss, comp);
975     log_forest();
976     result.log_forest();
977     return result;
978   }
979
980   virtual SplayTree split_left(Node *other_root) {
981     if (other_root == nullptr)
982       return SplayTree(nullptr, oss, comp);
983     replace(other_root, other_root->right);
984     other_root->parent = nullptr;
985     other_root->right = nullptr;
986     SplayTree result = SplayTree(other_root, oss, comp);
987     log_forest();
988     result.log_forest();
989     return result;
990   }
991
992 public:
993   virtual SplayTree split_after(const T &x) {
994     log("Split after:\n");
995     return split_right(ordered_after(x));
996   }
997
998   virtual SplayTree split_not_before(const T &x) {
999     log("Split not before:\n");
1000    return split_right(ordered_not_before(x));
1001  }
1002
1003  virtual SplayTree split_before(const T &x) {
1004    log("Split before:\n");
1005    return split_left(ordered_before(x));
1006  }
1007
1008  virtual SplayTree split_not_after(const T &x) {
1009    log("Split not after:\n");
1010    return split_left(ordered_not_after(x));
1011  }
1012
1013  // O(h) worst case, O(log n) amortized
1014  virtual std::pair<Node *, bool> top_down_search(const T &x) const {
1015    if (root == nullptr)
1016      return {nullptr, false};
1017    log("Top-down search:\n");
1018    bool found = false;
1019    Node *old_parent = root->parent;
1020    signed char old_side = is_left_child(root);
1021    root->parent = nullptr;
1022    Node *L = nullptr;
1023    Node *l = nullptr;
```

```
    Node *R = nullptr;
    Node *r = nullptr;
    while (true) {
      if (comp(x, root->value)) {
        if (root->left == nullptr)
          break;
        if (comp(x, root->left->value)) {
          log("LL zig-zig step:\n");
          root = rotate_right(root);
          if (root->left == nullptr)
            break;
        } else
          log("LR zig-zag step:\n");
        log("Link right:\n");
        if (R != nullptr) {
          r->left = root;
          root->parent = r;
          r = r->left;
        } else {
          R = root;
          r = R;
        }
        root = root->left;
        root->parent->left = nullptr;
        root->parent = nullptr;
        log("Right tree:");
        log_forest(R);
      } else if (comp(root->value, x)) {
        if (root->right == nullptr)
          break;
        if (comp(root->right->value, x)) {
          log("RR zig-zig step:\n");
          root = rotate_left(root);
          if (root->right == nullptr)
            break;
        } else
          log("RL zig-zag step:\n");
        log("Link left:\n");
        if (L != nullptr) {
          l->right = root;
          root->parent = l;
          l = l->right;
        } else {
          L = root;
          l = L;
        }
        root = root->right;
        root->parent->right = nullptr;
        root->parent = nullptr;
        log("Left tree:");
        log_forest(L);
      } else {
        found = true;
        break;
      }
    }
    if (L != nullptr) {
      if (root->left != nullptr) {
        l->right = root->left;
        root->left->parent = l;
        log("Left tree:");
        log_forest(L);
```

```
      }
      root->left = L;
      L->parent = root;
    }
    if (R != nullptr) {
      if (root->right != nullptr) {
        r->left = root->right;
        root->right->parent = r;
        log("Right tree:");
        log_forest(R);
      }
      root->right = R;
      R->parent = root;
    }
    root->parent = old_parent;
    switch (old_side) {
    case 1:
      old_parent->left = root;
      break;
    case 0:
      old_parent->right = root;
      break;
    }
    log("Link back:");
    log_forest();
    return {root, found};
  }

  // O(h) worst case, O(log n) amortized
  virtual Node *top_down_find_min() const {
    if (root == nullptr)
      return nullptr;
    log("Top-down find min:\n");
    Node *old_parent = root->parent;
    signed char old_side = is_left_child(root);
    root->parent = nullptr;
    Node *R = nullptr;
    Node *r = nullptr;
    while (true) {
      if (root->left == nullptr)
        break;
      log("LL zig-zig step:\n");
      root = rotate_right(root);
      if (root->left == nullptr)
        break;
      log("Link right:\n");
      if (R != nullptr) {
        r->left = root;
        root->parent = r;
        r = r->left;
      } else {
        R = root;
        r = R;
      }
      root = root->left;
      root->parent->left = nullptr;
      root->parent = nullptr;
      log("Right tree:");
      log_forest(R);
    }
    if (R != nullptr) {
      if (root->right != nullptr) {
```

```
      r->left = root->right;
      root->right->parent = r;
      log("Right tree:");
      log_forest(R);
    }
    root->right = R;
    R->parent = root;
  }
  root->parent = old_parent;
  switch (old_side) {
  case 1:
    old_parent->left = root;
    break;
  case 0:
    old_parent->right = root;
    break;
  }
  log("Link back:");
  log_forest();
  return root;
}

// O(h) worst case, O(log n) amortized
virtual Node *top_down_find_max() const {
  if (root == nullptr)
    return nullptr;
  log("Top-down find max:\n");
  Node *old_parent = root->parent;
  signed char old_side = is_left_child(root);
  root->parent = nullptr;
  Node *L = nullptr;
  Node *l = nullptr;
  while (true) {
    if (root->right == nullptr)
      break;
    log("RR zig-zig step:\n");
    root = rotate_left(root);
    if (root->right == nullptr)
      break;
    log("Link left:\n");
    if (L != nullptr) {
      l->right = root;
      root->parent = l;
      l = l->right;
    } else {
      L = root;
      l = L;
    }
    root = root->right;
    root->parent->right = nullptr;
    root->parent = nullptr;
    log("Left tree:");
    log_forest(L);
  }
  if (L != nullptr) {
    if (root->left != nullptr) {
      l->right = root->left;
      root->left->parent = l;
      log("Left tree:");
      log_forest(L);
    }
    root->left = L;
```

```
      L->parent = root;
    }
    root->parent = old_parent;
    switch (old_side) {
    case 1:
      old_parent->left = root;
      break;
    case 0:
      old_parent->right = root;
      break;
    }
    log("Link back:");
    log_forest();
    return root;
  }

  // Return node, inserted
  // O(h)
  virtual std::pair<Node *, bool> top_down_insert(const T &x) {
    if (root == nullptr) {
      root = new Node(x);
      log("Top-down insert:\n");
      log("Insert:");
      log_forest();
      return {root, true};
    }
    log("Top-down insert:\n");
    auto [node, found] = top_down_search(x);
    if (found)
      return {root, false};
    root = new Node(x);
    if (comp(x, node->value)) {
      root->left = node->left;
      node->left->parent = root;
      node->left = nullptr;
      root->right = node;
    } else {
      root->right = node->right;
      node->right->parent = root;
      node->right = nullptr;
      root->left = node;
    }
    root->parent = node->parent;
    switch (is_left_child(node)) {
    case 1:
      node->parent->left = root;
      break;
    case 0:
      node->parent->right = root;
      break;
    }
    node->parent = root;
    log("Insert:");
    log_forest();
    return {root, true};
  }

  virtual void top_down_join_right(Node *other) {
    if (other == nullptr)
      return;
    log("Top-down join right:\n");
    log("To join:");
```

```
    log_forest(other);
    if (root == nullptr) {
      root = other;
      other->parent = nullptr;
    } else {
      top_down_find_max();
      root->right = other;
      other->parent = root;
    }
    log("Join:");
    log_forest();
  }

  virtual void top_down_join_left(Node *other) {
    if (other == nullptr)
      return;
    log("Top-dowm join left:\n");
    log("To join:");
    log_forest(other);
    if (root == nullptr) {
      root = other;
      other->parent = nullptr;
    } else {
      top_down_find_min();
      root->left = other;
      other->parent = root;
    }
    log("Join:");
    log_forest();
  }

  // O(h) worst case, O(log n) amortized
  virtual std::pair<Node *, bool> top_down_remove(const T &x) {
    log("Top-down delete:\n");
    auto [victim, found] = top_down_search(x);
    if (found == false)
      return {victim, false};
    log("Delete:\n");
    Node *right = root->right;
    if (root->left != nullptr) {
      replace(root, root->left, true);
      if (right != nullptr)
        right->parent = nullptr;
      top_down_join_right(right);
    } else {
      replace(root, right, true);
      log_forest();
    }
    return {nullptr, true};
  }
};
```